# Kavana

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# Multi-Modal Stress and Cognitive Workload Assessment




## PROJECT REPORT

*Submitted by*

Group Number: 50

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Kavana M A (1BM23CS145)
Khushi Nataraj (1BM23CS151)

3 *in partial fulfillment for the award of the degree of*

**BACHELOR OF ENGINEERING**

in

**COMPUTER SCIENCE AND ENGINEERING**

Under the Guidance of

**Dr. Selva Kumar S**

11 Professor & Head, Department of CSE, BMSCE




**B.M.S. COLLEGE OF ENGINEERING**
(Autonomous Institution under VTU)
BENGALURU - 560019

Academic Year - 2025–2026

# Certificate

This is to certify that the project work entitled **"Multi-Modal Stress and Cognitive Workload Assessment"** carried out by **Ikshitha P (1BM23CS118), Jesvita Dias (1BM23CS126), Kavana M A (1BM23CS145) and Khushi Nataraj (1BM23CS151)**, who are bonafide students of B.M.S. College of Engineering, is submitted in partial fulfilment for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi, during the academic year 2025–2026. The project report has been approved as it satisfies the academic requirements in respect of Major Project (23CS6PWPP1) prescribed for the said degree.

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Examiner 1

Examiner 2

# Abstract

**Domain.** Cognitive workload assessment is central to human–computer interaction, adaptive learning, clinical diagnosis and operator-performance monitoring in safety-critical fields such as aviation and healthcare. Brain-based sensing, in particular electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS), allows workload to be measured objectively rather than inferred from self-report.

**Problem and research gap.** Existing practice relies on subjective questionnaires and behavioural analysis, which are biased and cannot be applied continuously. Single-modality EEG systems are sensitive to eye-blink and muscle artefacts and carry no haemodynamic information, while single-modality fNIRS responds slowly. Prior EEG–fNIRS studies often use classical classifiers or subject-dependent evaluation, so their reported accuracy says little about how a model behaves on a person it has never seen. Few studies compare fusion strategies under strict subject-independent evaluation.

**Proposed solution.** This project develops a multimodal deep-learning framework that fuses EEG and fNIRS to classify cognitive workload as LOW (0-back) or HIGH (2-back, 3-back). EEG is encoded by a CNN–BiLSTM with temporal attention pooling, fNIRS (HbO and HbR) by a temporal CNN, and the two 128-dimensional representations are fused before classification.

**Implementation.** The public TU Berlin simultaneous EEG–fNIRS n-back dataset (24 subjects, 648 paired 42-second epochs) was band-pass filtered, mean-removed, common-average referenced (EEG) and baseline-corrected (fNIRS). Models were implemented in PyTorch with MNE-Python and scikit-learn, trained with class-weighted cross-entropy, and evaluated with a strict 24-fold Leave-One-Subject-Out (LOSO) protocol using a separate validation subject; all normalisation statistics came from training subjects only.

**Results and conclusion.** On unseen subjects the final representation-level model achieved 66.36% accuracy, 57.70% Macro-F1 and 0.7842 ROC-AUC, improving on EEG-only and fNIRS-only baselines. Simple 50:50 decision-level fusion gave the best Macro-F1 (62.26%). These scores are lower than many published figures, mainly because of strict subject-independent evaluation, a small sample and class imbalance. The work provides a leakage-safe benchmark of fusion strategies and a basis for future real-time brain–computer interface applications.

# DECLARATION

We, hereby declare that the Major Project Phase-2 work entitled **"Multi-Modal Stress and Cognitive Workload Assessment"** is a bonafide work and has been carried out by us under the guidance of Dr. Selva Kumar S, Professor & Head, Department of Computer Science and Engineering, B.M.S. College of Engineering, Bengaluru, in partial fulfillment of the requirements of the degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi. We further declare that, to the best of our knowledge and belief, this project has not been submitted either in part or in full to any other university for the award of any degree.

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**Place:** Bengaluru **Date:** _____

Certified that these candidates are students of the Computer Science and Engineering Department of B.M.S. College of Engineering. They have carried out the project work titled "Multi-Modal Stress and Cognitive Workload Assessment" as Major Project Phase-2 work, in partial fulfilment of the requirement for the award of the B.E. degree by VTU. The work is original and duly certified.

**Guide Name:** Dr. Selva Kumar S **Date:** _____

**Signature:**

# Acknowledgment

We would like to express our deepest gratitude to Dr. Bheemsha Arya, Principal of B.M.S. College of Engineering, and Dr. Kavitha Sooda, Head of the Department of Computer Science and Engineering, for providing the facilities and an enabling atmosphere for research and development.

We thank Dr. Selva Kumar S, Professor & Head of the Department of Computer Science and Engineering, BMS College of Engineering, for his valuable inputs throughout the project. His expertise in deep learning, brain-signal analysis and intelligent healthcare systems played an important role in the success of this study.

Lastly, we acknowledge the contribution of the other faculty and staff members of the Department of Computer Science and Engineering in developing this project.

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# Chapter 1 Introduction

## 1.1 Background of the Study

Modern work and study environments place a heavy load on human attention and memory. When demands exceed capacity, performance and psychological well-being suffer. *Cognitive workload* is the mental effort a person invests to meet task demands, and measuring it objectively is useful in education, healthcare, aviation, driving and human–machine systems.

Two non-invasive brain-sensing technologies are widely used. **Electroencephalography (EEG)** records the brain's electrical activity at millisecond resolution [1, 2]. **Functional Near-Infrared Spectroscopy (fNIRS)** uses near-infrared light to track changes in oxygenated (HbO) and deoxygenated (HbR) haemoglobin, a slow metabolic response linked to neural activity [3, 9]. This project combines both with hybrid neural networks [4, 5, 6] to learn the relationship between fast neuro-electrical activity and the slower haemodynamic response, and uses it to classify workload on unseen subjects.

## 1.2 Research Motivation

- **Artefact sensitivity:** EEG-only systems are strongly affected by eye-blink and muscle artefacts [7, 8], which lowers the signal-to-noise ratio (SNR) until brain activity is hard to separate from movement.
- **Latent fatigue detection:** Mental fatigue has a metabolic component. EEG captures immediate attentional change but not the cost of prolonged effort; fNIRS adds the haemodynamic information [3, 9] needed to detect fatigue before it causes errors.
- **Generalisation:** Many published systems are tuned per person. A model that works on new users without calibration is far more practical [15, 17].

## 1.3 Objective

The goal is a hardware-agnostic software framework that classifies mental workload from EEG and fNIRS more reliably than unimodal frameworks. The specific objectives are:

1. Build a pipeline that pairs millisecond-level EEG with second-level fNIRS using task event markers (TTL/software triggers in future acquisition).
2. Use standardised workload paradigms (N-back; Stroop and Corsi Block planned with PsychoPy [10]) that produce measurable changes in working memory and inhibitory control [11, 12].

3. Design CNN–BiLSTM–attention encoders and compare unimodal, decision-level and representation-level fusion.

4. Establish Macro-F1, Accuracy and ROC-AUC benchmarks under subject-independent (LOSO) evaluation [13, 14].

## 1.4 Scope

This phase covers computational modelling and comparative analysis. Because fNIRS hardware is expensive and unavailable, the scope is:

- **Secondary data analysis** using the public TU Berlin simultaneous EEG–fNIRS n-back dataset [15, 16].
- **Subject-independent training and evaluation** on completely unseen subjects, to reduce calibration needs [2, 5].
- **Binary classification** (LOW vs HIGH workload). Three-state classification, real-time acquisition, dashboard and API integration are outside this phase (see Chapter 7).
- **A PsychoPy set-up** [10] prepared as a plug-and-play environment for future hardware work.

## 1.5 SDG Justification

**SDG 3 – Good Health and Well-being.** Workload, stress and fatigue monitoring can support preventive care in high-stress settings such as healthcare, aviation and emergency response.

**SDG 9 – Industry, Innovation and Infrastructure.** The work advances AI, deep learning and brain–computer interfaces, and contributes to infrastructure for cognitive analytics.

**SDG 17 – Partnerships for the Goals.** It relies on open datasets, open-source tools and cross-disciplinary collaboration between neuroscience, biomedical engineering and AI.

## 1.6 Existing System

Current approaches use subjective questionnaires and behavioural measures, or single/multi-channel EEG with classical classifiers such as SVM and LDA [13]. Their limitations are:

- EEG cannot localise activity in the dorsolateral prefrontal cortex (dlPFC) and misses the metabolic cost of sustained effort.
- Questionnaires are biased and cannot run continuously [1].

- Fixed thresholds ignore inter-subject variability, giving poor generalisation [2, 17].
- Classical classifiers need hand-crafted features and cannot learn cross-modal relationships [13, 23].

## 1.7 Proposed System

The proposed system is a multimodal deep-learning pipeline with the following levels.

- **Data level:** paired EEG (electrical) and fNIRS (metabolic) epochs from the TU Berlin dataset.
- **Processing level:** band-pass filtering, mean/baseline removal and common-average referencing. ICA [8] and the Modified Beer–Lambert Law [18] belong to a full raw-data pipeline and are not applied to the already-converted dataset.
- **Feature/fusion level:** a CNN extracts local patterns from EEG [6], a BiLSTM models long-range dependencies [19], temporal attention [21] weights informative time steps, and a temporal CNN encodes fNIRS; the learned representations are fused.
- **Classification level:** Softmax output for LOW/HIGH, extendable to Rest, Optimal Flow and Cognitive Saturation.

| Component | Technical implementation |
|---|---|
| Experimental tool | PsychoPy [10] for millisecond-accurate stimulation (future acquisition) |
| Cognitive tasks | N-back (used in dataset); Stroop and Corsi Block (planned) |
| Synchronisation | Event-marker (TTL-style) alignment of different sampling rates |
| Features | EEG: learned temporal representations (PSD/band power planned [14]); fNIRS: HbO, HbR |
| Model | CNN–BiLSTM with attention for EEG; temporal CNN for fNIRS; fusion MLP |
| Evaluation | Macro-F1 (primary), Accuracy, ROC-AUC, confusion matrix |

Table 1.1: Technical implementation summary

## 1.8 Work Plan

The project ran in two phases over the academic year 2025–26: Phase-1 (study, design, dataset selection) and Phase-2 (implementation, experiments, analysis, documentation). Table 1.2 lists the activities, the deliverable of each and its status. **[Replace the week ranges with your actual calendar dates.]**

Table 1.2: Work plan

| Sl. | Activity | Timeline | Deliverable / outcome | Status |
|---|---|---|---|---|
| 1 | Problem definition, literature survey of 26 references | Phase-1, Weeks 1–4 | Research gaps and objective mapping (Ch. 2) | Completed |
| 2 | Requirement analysis, feasibility, tool selection | Phase-1, Weeks 4–6 | SRS, tool-chain, cost estimate (Ch. 3) | Completed |
| 3 | Dataset selection (TU Berlin EEG–fNIRS n-back) and study of recording protocol | Phase-1, Weeks 6–8 | Dataset summary, labelling rule (0-back $\rightarrow$ LOW; 2/3-back $\rightarrow$ HIGH) | Completed |
| 4 | System design: architecture, modules, use case, class and sequence design | Phase-1, Weeks 8–10 | Design documents (Ch. 4) | Completed |
| 5 | Preprocessing and epoching (filtering, CAR, baseline removal) | Phase-2, Weeks 1–3 | 648 paired epochs: EEG $28 \times 8400$, fNIRS $36 \times 420 \times 2$ | Completed |
| 6 | LOSO framework and leakage-safe normalisation | Phase-2, Weeks 3–4 | 24-fold split (22/1/1 subjects) | Completed |
| 7 | Unimodal EEG (CNN–BiLSTM) and fNIRS (CNN) baselines | Phase-2, Weeks 4–6 | Baseline metrics (Table 6.2) | Completed |
| 8 | Fusion models: representation-level, decision-level (50:50, thresholded), learned and logistic fusion | Phase-2, Weeks 6–9 | Comparison of fusion variants (Table 6.3) | Completed |

| Sl. | Activity | Timeline | Deliverable / outcome | Status |
|---|---|---|---|---|
| 9 | Temporal attention for EEG, hyper-parameter and dropout selection | Phase-2, Weeks 8–10 | Final model | Completed |
| 10 | Result analysis, comparison with literature, error analysis | Phase-2, Weeks 10–12 | Chapter 6 | Completed |
| 11 | Report writing, review corrections, similarity check | Phase-2, Weeks 12–14 | Final report | Completed |
| 12 | Hybrid features, LOW-class recall, calibration, ICA/motion correction, three-state classification | Next phase | Improved Macro-F1 (target $\geq 0.80$) | Future work |
| 13 | Real-time acquisition with PsychoPy and hardware, dashboard and API integration | Next phase | Real-time prototype ($<$ 200 ms latency) | Future work |

# Chapter 2 Literature Survey

## 2.1 Overview

Cognitive Workload Assessment (CWA) has grown within neuroscience, biomedical engineering and AI because intelligent systems increasingly need real-time monitoring of cognitive state. Workload was traditionally measured with questionnaires and performance measures, which are useful but subjective and not continuous. Researchers therefore moved to physiological sensing.

EEG offers high temporal resolution and reflects attention, memory and stress through theta, alpha and beta activity [1, 2, 11, 14], but it is corrupted by blinks, muscle activity and movement [7, 8]. fNIRS measures haemodynamic changes (HbO/HbR) in the prefrontal cortex, giving metabolic information and better localisation [3, 9], but is slow. Combining them exploits neurovascular coupling and improves robustness [1, 2, 3]. Deep models (CNN, LSTM, CNN–LSTM) now extract spatio-temporal features from such data [4, 5, 6, 19], and attention-based fusion is emerging [21, 22].

## 2.2 Detailed Survey

### 2.2.1 EEG-based workload assessment

Gevins and Smith [11] showed that EEG spectral activity tracks workload and attention, which established theta and alpha bands as workload indicators. Welch's method [14] for power spectral density (PSD) became the standard frequency-domain tool. Gramfort et al. [7] released MNE-Python for filtering, epoching and artefact visualisation, and Hyvärinen and Oja [8] introduced ICA to separate neural signals from ocular and muscle artefacts. LeCun et al. [6] laid the foundation for CNN-based hierarchical feature learning. Kingphai and Moshfeghi [20] showed that deep networks outperform traditional machine learning for EEG workload prediction, and a review of deep approaches [5] highlights CNN, RNN and hybrid models. Sharma and Gupta [17] studied correlation-based channel selection for EEG workload classification.

### 2.2.2 fNIRS-based workload assessment

24

Ayaz et al. [3] used optical brain monitoring for operator training and found prefrontal HbO changes correlate with workload. Boas et al. [18] described diffuse optical tomography and the Modified Beer–Lambert Law that converts optical density into haemoglobin concentration. Abibullaev and An [13] classified workloads from fNIRS using wavelet features and SVM, confirming the value of HbO/HbR. Ho et al. [25] applied deep learning to multi-channel fNIRS

to discriminate workload levels, and a Frontiers study [9] reinforced fNIRS as an indicator of sustained mental workload. Vourvopoulos and Liarokapis [24] linked cognitive load during an emergency simulation game to prefrontal activation measured by fNIRS.

### 2.2.3 Multimodal EEG–fNIRS fusion

Hirachan et al. [1] built a multimodal sensing system showing improved robustness. Putze et al. [2] reviewed fusion techniques and reported that EEG–fNIRS fusion outperforms single modalities. Khan and Hasan [4] used a CNN–LSTM with decision-level fusion to classify cognitive tasks. Al-Shargie et al. [23] applied decision-level fusion to stress assessment with fewer errors. Asif et al. presented CLARE [22], a real-time multimodal load-assessment framework, and a further multimodal EEG–fNIRS study is listed as [26].

### 2.2.4 Sequence models, attention and tools

Hochreiter and Schmidhuber [19] introduced LSTM networks that learn long-term temporal dependencies. Vaswani et al. [21] introduced attention, now used for cross-modal fusion. Peirce [10] developed PsychoPy for millisecond-accurate experiment timing, and the defence-science study [12] used standardised tasks such as N-back with neurophysiological signals to measure load.

### 2.2.5 Datasets

15 Jo et al. [15] released MOCAS, a multimodal dataset for workload on simultaneous tasks, and Shin et al. [16] released the simultaneous EEG–NIRS open dataset on which this project is built. Such datasets make subject-independent evaluation possible.

### 2.2.6 Comparative analysis of the referred papers

Table 2.1 compares every paper cited in this report. The "limitation" column states what the work does not address relative to this project's aim of subject-independent, fusion-comparison evaluation.

Table 2.1: Comparative analysis of referred literature

| Ref | Authors, year | Modality / type | Method and key finding | Limitation / difference from this work |
|---|---|---|---|---|
| [1] | Hirachan et al., 2022 | EEG + fNIRS sensors | Multimodal workload measurement; better robustness and reliability | No controlled comparison of fusion strategies under LOSO |

| Ref | Authors, year | Modality / type | Method and key finding | Limitation / difference from this work |
|---|---|---|---|---|
| [2] | Putze et al., 2020 | Review (multimodal) | EEG–fNIRS fusion outperforms single modality; discusses calibration issues | Review only; no experiments on a common protocol |
| [3] | Ayaz et al., 2012 | fNIRS | Prefrontal HbO rises with workload; operator training | Single modality, slow response, no deep model |
| [4] | Khan and Hasan, 2023 | EEG + fNIRS | CNN–LSTM with decision-level fusion for cognitive tasks | Different tasks; subject independence and per-class recall not the focus |
| [5] | Review, PMC, 2024 | EEG (deep learning review) | CNN, RNN and hybrids extract workload features effectively | Survey; no fNIRS fusion experiment |
| [6] | LeCun et al., 2015 | Deep learning theory | Hierarchical feature learning with CNNs | General theory, not workload-specific |
| [7] | Gramfort et al., 2014 | EEG/MEG software | MNE: filtering, epoching, visualisation | Tool, not a classifier |
| [8] | Hyvärinen and Oja, 2000 | Signal processing | ICA separates neural activity from EOG/EMG artefacts | Needs raw data; not applied in this project's dataset pipeline |
| [9] | Frontiers Hum. Neurosci., 2022 | fNIRS | fNIRS-based workload assessment through haemodynamics | Single modality |
| [10] | Peirce, 2007 | Experiment software | PsychoPy, millisecond-precise stimulus timing | Stimulus delivery only; used here for planned acquisition |
| [11] | Gevins and Smith, 2003 | EEG | Spectral EEG indices track workload; theta/alpha bands | No fNIRS or deep fusion |
| [12] | Defence Life Sci. J., 2020 | Neurophysiological signals | Standardised tasks such as N-back for workload | Descriptive; no multimodal deep learning |
| [13] | Abibullaev and An, 2012 | fNIRS | Wavelet features + SVM; HbO/HbR informative | Classical classifier, handcrafted features, no EEG |
| [14] | Welch, 1967 | Spectral analysis | PSD estimation via averaged FFT | Planned as future hybrid feature |
| [15] | Jo et al., 2025 | Dataset (MOCAS) | Multimodal data for objective workload on simultaneous tasks | Dataset paper; shows poor cross-subject generalisability |
| [16] | Shin et al., 2018 | Dataset (EEG + NIRS) | Open simultaneous n-back data; used in this project | Small number of subjects and blocks |
| [17] | Sharma and Gupta, 2025 | EEG | Correlation-based channel selection for workload | EEG only; channel selection not used here |

| Ref | Authors, year | Modality / type | Method and key finding | Limitation / difference from this work |
|---|---|---|---|---|
| [18] | Boas et al., 2004 | fNIRS physics | Diffuse optical tomography; Modified Beer–Lambert Law | Theory; dataset already provides HbO/HbR |
| [19] | Hochreiter and Schmidhuber, 1997 | Sequence model | LSTM learns long-term dependencies | Generic; BiLSTM variant used for EEG here |
| [20] | Kingphai and Moshfeghi, 2024 | EEG | Deep models beat traditional ML for workload | EEG only, no haemodynamic information |
| [21] | Vaswani et al., 2017 | Attention | Attention and transformers; basis of cross-modal fusion | NLP origin; adapted as temporal attention pooling |
| [22] | Asif et al. (CLARE), 2025 | Multimodal, real time | Real-time load assessment, decision-level learning | Real-time focus; ours is offline with real-time as future work |
| [23] | Al-Shargie et al., 2017 | EEG + fNIRS | Decision-level fusion lowers error in stress assessment | Stress task; classical features |
| [24] | Vourvopoulos and Liarokapis, 2022 | fNIRS | Cognitive load in emergency game linked to prefrontal activation | Single modality, game-based task |
| [25] | Ho et al., 2019 | fNIRS (multi-channel) | Deep learning for workload levels | Single modality; no EEG fusion |
| [26] | IEEE Xplore, 2024 | EEG + fNIRS | Multimodal workload assessment with fusion | Details to be confirmed from source; fusion not compared under LOSO |

**Observations.** (i) By modality, EEG papers [11, 17, 20] give temporal detail but are artefact-prone, fNIRS papers [3, 13, 25] give metabolic information but respond slowly, and the multimodal papers [1, 2, 4, 22, 23, 26] report that fusion helps. (ii) By technique, older work uses SVM/LDA with handcrafted features [13, 23], newer work uses CNN/LSTM [4, 20, 25], and attention is still rarely used for EEG–fNIRS [21]. (iii) By fusion level, most fusion studies use decision-level fusion [4, 23, 22]; representation-level fusion with attention is less explored. (iv) By evaluation, the cited works rarely stress strict leave-one-subject-out testing with a separate validation subject, and the dataset papers [15, 17] note poor cross-subject generalisation.

Table 2.2 places this work against the main multimodal studies.

## 2.3 Research Gaps

- **Artefact sensitivity of EEG:** low SNR and EOG/EMG/movement corruption [7, 8].

| | [1] | [4] | [23] | **This work** |
|---|---|---|---|---|
| Modality | EEG+fNIRS | EEG+fNIRS | EEG+fNIRS | EEG+fNIRS |
| Classifier | Multimodal system | CNN–LSTM | Decision fusion | CNN–BiLSTM + attention, CNN |
| Fusion level | Sensor level | Decision | Decision | Representation and decision (compared) |
| Task | Workload | Cognitive tasks | Stress | n-back workload |
| Attention | No | No | No | Temporal attention |
| Evaluation | Not LOSO-focused | Not LOSO-focused | Not LOSO-focused | 24-fold LOSO + validation subject |

Table 2.2: Comparison with representative multimodal studies

- **No metabolic information in unimodal systems** [3, 9].
- **Time alignment:** millisecond EEG versus second-scale fNIRS [1, 2].
- **Reliance on classical ML** requiring feature engineering [13, 23].
- **Poor cross-subject generalisation** [15, 17].
- **Mostly offline studies;** little real-time monitoring [22].
- **Limited attention-based multimodal fusion** in a subject-independent setting [21].
- **Few scalable, hardware-independent frameworks** for medical, educational and practical use [2, 22].

## 2.4 Mapping of Project Objectives with Research Gaps

Table 2.3: Gap mapping

| **Research gap** | **Proposed objective** |
|---|---|
| Artefacts (EOG, EMG, motion) in EEG [7, 8] | Preprocessing pipeline with filtering, referencing and baseline/mean removal (ICA planned) |
| No metabolic measurement in unimodal systems [3, 9] | Combine EEG and fNIRS (HbO/HbR) |
| Alignment of EEG and fNIRS [1, 2] | Event-marker alignment into 42-second paired epochs |

| | |
|---|---|
| Classical ML, manual features [13, 23] | Deep CNN and BiLSTM encoders with fusion |
| Generalisation across subjects [15, 17] | Public dataset with Leave-One-Subject-Out evaluation |
| Offline-only studies [22] | Offline evaluation now; architecture designed for later real-time use |
| Little attention-based fusion [21] | Temporal attention in the EEG encoder inside representation-level fusion |
| Hardware-dependent, unscalable systems [2, 22] | Software-based, hardware-agnostic framework |

# Chapter 3 Requirement Analysis and Specification

## 3.1 Stakeholders

| Stakeholder | Interest |
|---|---|
| Researcher / administrator | Loads data, runs experiments, reads metrics and reports |
| Subject / user | Provides EEG–fNIRS data; benefits from workload feedback (future real-time system) |
| Guide and evaluators | Review technical soundness and reproducibility |
| Application developers | Future consumers of workload predictions via API |
| Domain users (education, healthcare, aviation) | Beneficiaries of workload monitoring |

## 3.2 Functional Requirements

- **FR1 Data acquisition:** load multi-channel EEG and fNIRS (HbO/HbR) recordings and event markers.
- **FR2 Preprocessing:** EEG band-pass 0.5–45 Hz, channel mean removal, common-average reference; fNIRS band-pass 0.01–0.5 Hz, baseline and residual mean removal.
- **FR3 Epoching and labelling:** cut 42-second epochs from task markers; label 0-back as LOW and 2/3-back as HIGH.
- **FR4 Splitting and normalisation:** create LOSO folds (22 train, 1 validation, 1 test subject); compute normalisation from training subjects only.
- **FR5 Representation learning:** produce 128-D EEG and 128-D fNIRS features with CNN-based encoders.
- **FR6 Fusion and classification:** fuse the representations and output LOW/HIGH with probabilities.
- **FR7 Evaluation:** report Accuracy, Macro-Precision/Recall/F1, ROC-AUC and confusion matrices per fold and pooled.

- **FR8 (future)** Hybrid features (PSD, HbO/HbR statistics), three-state output, real-time streaming.

## 3.3 Non-Functional Requirements

- **Accuracy:** maximise Macro-F1 under subject-independent testing. The initial target $F1 \geq 0.80$ was not reached in this phase (achieved $\approx 0.58$).
- **Latency (future):** classification in under 200 ms for real-time feedback [22].
- **Scalability and portability:** hardware-agnostic (OpenBCI, Muse, g.tec).
- **Robustness:** tolerate non-stationary signals [7].
- **Subject independence:** no information from the test subject may be used for normalisation, tuning or model selection.
- **Reproducibility:** identical preprocessing, 24-fold LOSO and random seed 42 for all experiments.

## 3.4 System Requirements

### 3.4.1 Hardware

| Component | Specification | Reason |
|---|---|---|
| CPU | Intel Core i7 (12th gen) | Multi-threaded epoch processing |
| GPU | NVIDIA RTX 3060 (6 GB) or better; Colab/Kaggle optional | Faster CNN–BiLSTM training over 24 folds; code runs on CPU or GPU automatically |
| RAM | 16 GB DDR4 minimum | Holds 648 epochs (EEG $28 \times 8400$, fNIRS $36 \times 420 \times 2$) |
| Storage | 512 GB SSD | Dataset, checkpoints and per-fold results |
| EEG sensor (future) | Dry electrodes, 8–16 channels | Real-time deployment; not used now |
| fNIRS sensor (future) | Dual-wavelength NIR source/detector | HbO/HbR from prefrontal cortex; not used now |

### 3.4.2 Software

OS: Windows 10/11 or Ubuntu 20.04 (recommended). Python 3.9+. PyTorch (CNN, BiLSTM, attention, fusion; class-weighted cross-entropy, Adam, ReduceLROnPlateau [6]). MNE-Python for loading, filtering, epoching and visualisation [7]. NumPy/SciPy/Pandas for filtering and tables. scikit-learn for metrics. PsychoPy for planned n-back/Stroop paradigms [10].

### 3.4.3 Cost estimation

All items (laptop/PC, open dataset, open-source software, free Colab/Kaggle, storage) are available at no cost: total **NIL**.

## 3.5 Data Requirements

Simultaneous EEG (28 channels, 200 Hz) and fNIRS (36 channels, 10 Hz, HbO and HbR) from the same subjects, with markers identifying 0-, 2- and 3-back blocks; at least 20 subjects so a subject-independent split is meaningful; labelled task conditions. The TU Berlin dataset [16] satisfies these (Section 5.2).

## 3.6 Constraints and Limitations

No fNIRS hardware is available, so only secondary data is used. Only 24 subjects and 27 epochs per subject exist. Training is limited to 12 epochs and runs on CPU/GPU chosen automatically, so extensive hyper-parameter search is not feasible. Labels are task conditions, not each person's measured effort.

## 3.7 Assumptions

(i) The public dataset is correctly synchronised and labelled. (ii) Task condition is a valid proxy for workload (0-back low, 2/3-back high). (iii) The subjects are representative enough for subject-independent evaluation. (iv) Results from this offline study indicate behaviour in a later real-time system.

## 3.8 Risks and Challenges

| Risk | Mitigation |
|---|---|
| Overfitting with 22 training subjects | Simple fNIRS CNN, dropout, early stopping |
| Class imbalance (216 LOW : 432 HIGH) | Class-weighted loss; Macro-F1 as primary metric |
| Information leakage | Normalisation from training subjects only; separate validation subject |
| High inter-subject variability | Report mean $\pm$ SD and pooled results; avoid over-interpreting 1–2 point gaps |
| Noisy EEG | Filtering and referencing now; ICA planned |
| Unreliable tuning on a single validation subject | Fixed small search space; multi-seed runs planned |

# Chapter 4 Designs

## 4.1 System Architecture Overview

The system is organised in layers (Figures 4.1 and 4.2). The **input layer** supplies EEG, fNIRS and task events. The **preprocessing and synchronisation layer** cleans each modality and aligns them by event markers. The **feature/representation layer** turns each modality into a 128-D vector (learned in the implemented system; spectral and haemodynamic statistics in the full design). The **fusion layer** combines the two vectors. The **classification layer** (fully connected, dropout, softmax) outputs LOW/HIGH with confidence. The **training layer** (loss, optimiser, scheduler, validation) and **evaluation layer** (metrics) close the loop, and the **output layer** (dashboard, reports, API) is reserved for future real-time use.

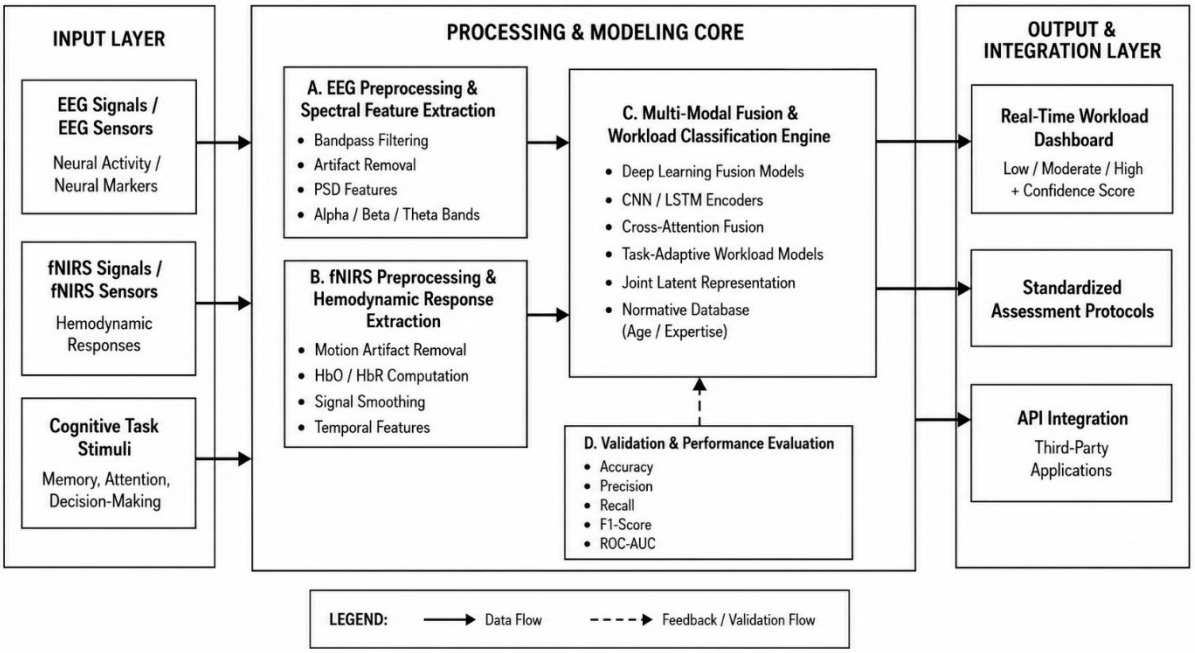



Figure 4.1: High-level design of multimodal signal processing, fusion and workload classification

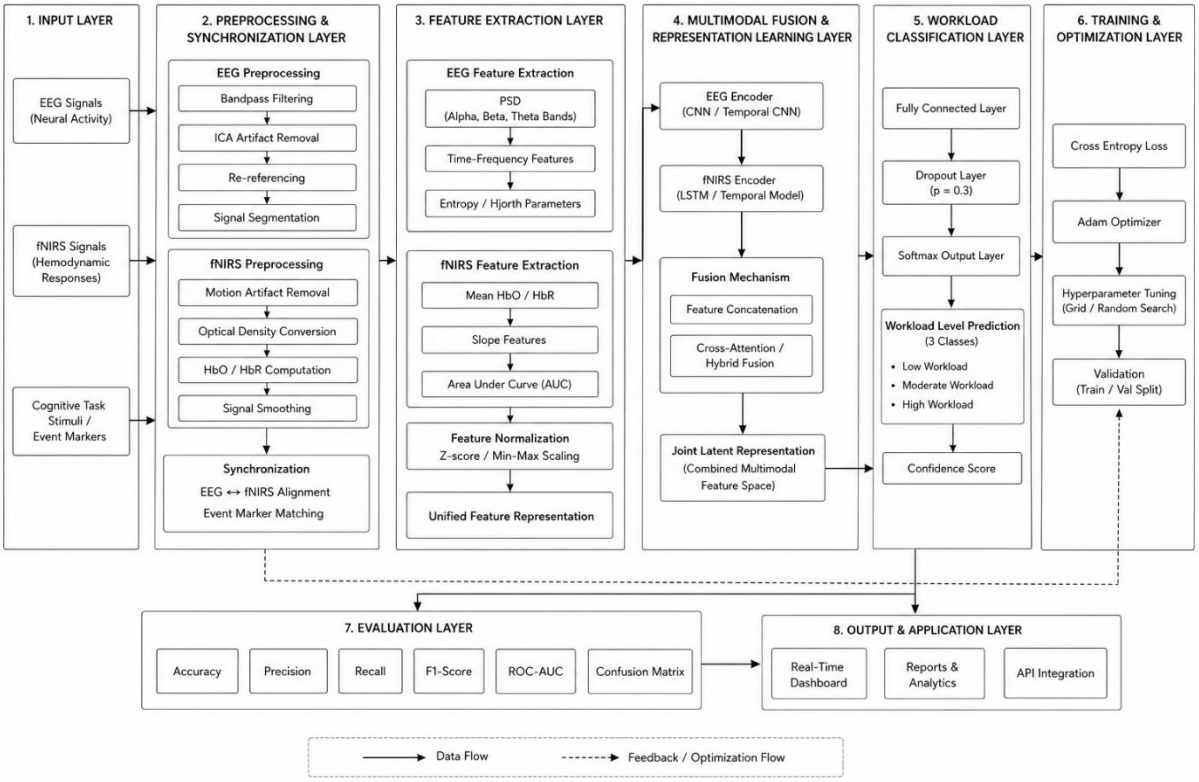



Figure 4.2: System architecture of the proposed system

## 4.2 Module/Component Design

Table 4.1: Module design

| Module | Responsibility | Input | Output |
|---|---|---|---|
| Dataset loader | Reads recordings and markers for 24 subjects | Raw files | Raw EEG, fNIRS, markers |
| Preprocessing module | Filtering, mean removal, CAR, baseline removal, epoching, labelling | Raw signals, markers | Paired epochs and labels |
| LOSO splitter | Builds 24 folds with 22/1/1 subjects | 648 epochs with subject IDs | Train, validation, test sets |
| Normaliser | Channel-wise $z$-score using training statistics only | Split sets | Normalised sets |

Table 4.1: Module design

| **Module** | **Responsibility** | **Input** | **Output** |
|---|---|---|---|
| EEG encoder | CNN $\rightarrow$ BiLSTM $\rightarrow$ attention pooling | $28 \times 8400$ | 128-D vector |
| fNIRS encoder | Three Conv1D blocks + global average pooling | $36 \times 420 \times 2$ (HbO/HbR as 2 channels) | 128-D vector |
| Fusion module | Concatenate, MLP $256 \rightarrow 128$, classifier $128 \rightarrow 2$ | Two 128-D vectors | Logits / probabilities |
| Trainer | Class-weighted loss, Adam, scheduler, early stopping, checkpoint | Train and validation data | Best checkpoint per fold |
| Evaluator | Metrics and confusion matrix | Test predictions | Per-fold and pooled results |

## 4.3 Detailed Design

### 4.3.1 Use-case design

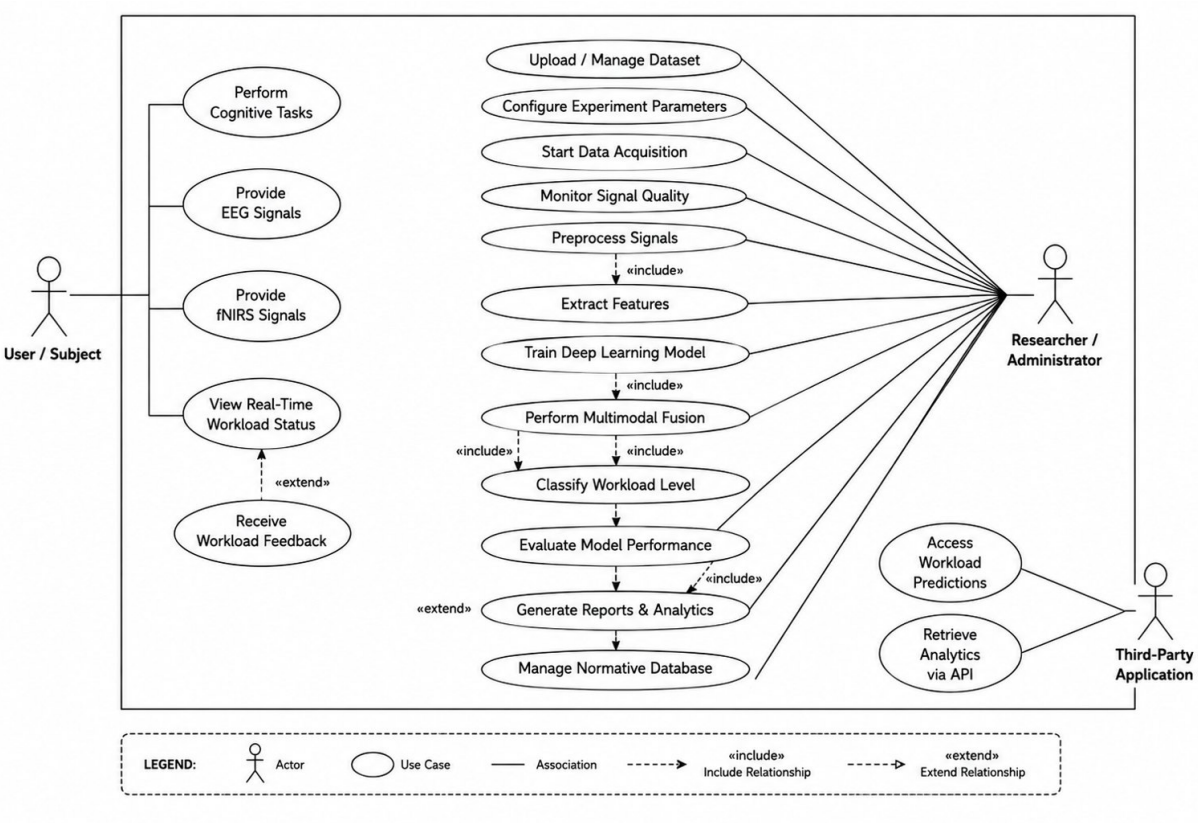



Figure 4.3: Use case diagram

Three actors interact with the system (Figure 4.3). The *subject* performs cognitive tasks, supplies EEG and fNIRS signals and, in the real-time design, views workload status and receives feedback. The *researcher/administrator* manages the whole experimental chain: uploading datasets, configuring experiments, starting acquisition, monitoring signal quality, preprocessing, extracting features, training, fusion, classification, evaluation, reporting and maintaining a normative database. A *third-party application* retrieves predictions and analytics through an API. Dependencies between use cases are significant: preprocessing is always *included* before feature extraction, fusion before classification, and training before evaluation; report generation *extends* evaluation and workload feedback *extends* status viewing, because they are optional add-ons. In the implemented offline prototype only the researcher's chain (load → preprocess → train → classify → evaluate) is realised.

### 4.3.2 Class design

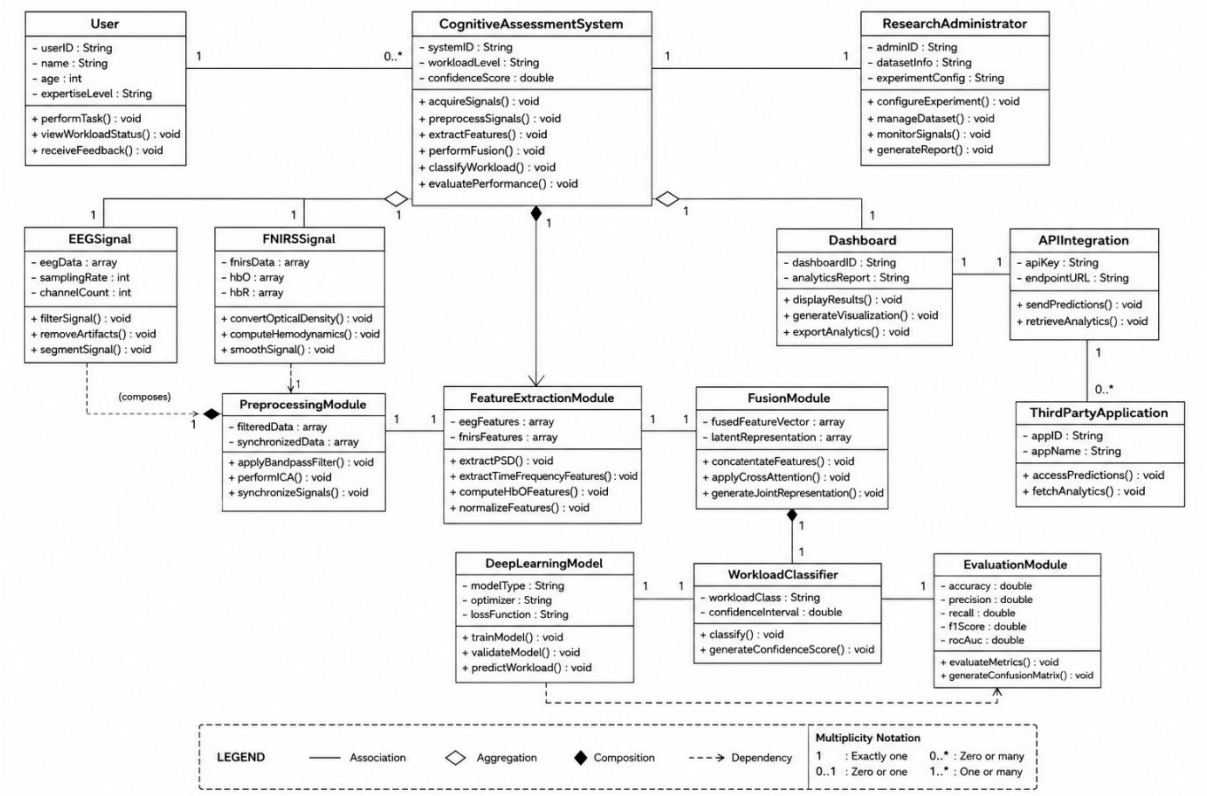



Figure 4.4: Class diagram

Figure 4.4 groups classes by responsibility. A central `CognitiveAssessmentSystem` controller owns the workflow and holds the current workload level and confidence. `EEGSignal` and `FNIRSSignal` hold raw arrays, sampling rate and channel count, and expose filtering, artefact removal, segmentation, optical-density conversion and smoothing. They are *aggregated* by the controller and *composed* into `PreprocessingModule`, which cannot exist without them and produces filtered, synchronised data. `FeatureExtractionModule` produces PSD/time-frequency features for EEG and HbO/HbR statistics for fNIRS and normalises them; `FusionModule` concatenates and optionally applies cross-attention to create a joint latent representation. `DeepLearningModel` trains and validates the network and predicts; `WorkloadClassifier` converts the representation into a class and confidence; `EvaluationModule` computes metrics. `Dashboard`, `APIIntegration` and `ThirdPartyApplication` expose results externally. In the implemented code these correspond to Python modules and PyTorch `nn.Module` classes: dataset loader, preprocessing functions, EEG encoder, fNIRS encoder, fusion head, training loop and metrics.

### 4.3.3 Sequence design

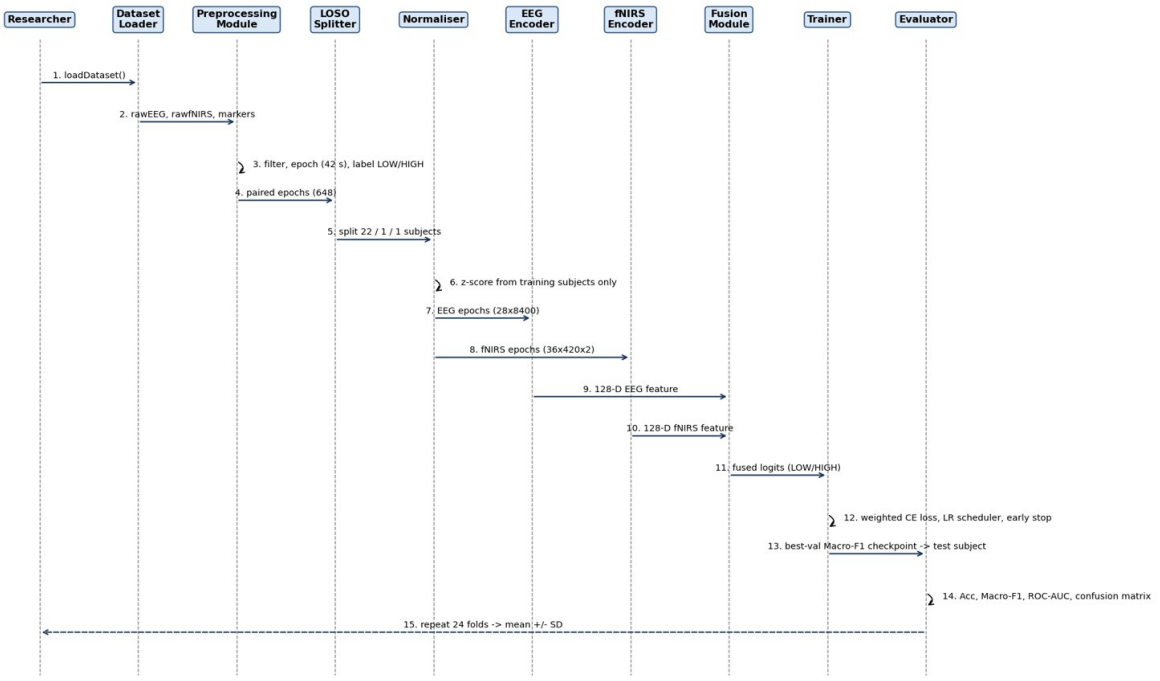



Figure 4.5: Sequence diagram of the offline LOSO pipeline

Figure 4.5 shows the ordered interaction between the Researcher, Dataset Loader, Preprocessing Module, LOSO Splitter, Normaliser, EEG Encoder, fNIRS Encoder, Fusion Module, Trainer and Evaluator:

1. The researcher calls `loadDataset()`.
2. The loader returns raw EEG, raw fNIRS and markers.
3. The preprocessing module filters and cuts 42-second epochs and labels them LOW/HIGH.
4. 648 paired epochs go to the LOSO splitter.
5. The splitter creates a 22/1/1 subject split for the current fold.
6. The normaliser computes $z$-score statistics from training subjects only.
7. EEG epochs ($28 \times 8400$) go to the EEG encoder and
8. fNIRS epochs ($36 \times 420 \times 2$) to the fNIRS encoder.
9. The EEG encoder returns a 128-D feature, and
10. the fNIRS encoder returns a 128-D feature to the fusion module.
11. The fusion module returns LOW/HIGH logits to the trainer.
12. The trainer minimises weighted cross-entropy, adjusts the learning rate and stops early.
13. The best validation-Macro-F1 checkpoint predicts the test subject.
14. The evaluator records Accuracy, Macro-F1, ROC-AUC and the confusion matrix.
15. Steps 5–14 repeat for all 24 folds, and mean $\pm$ SD is returned to the researcher.

## 4.4 Data Design

| Data item | Design |
|---|---|
| EEG epoch | Float tensor $28 \times 8400$ (channels $\times$ samples, 42 s at 200 Hz) |
| fNIRS epoch | Float tensor $36 \times 420 \times 2$ (channels $\times$ samples $\times$ {HbO, HbR}; 42 s at 10 Hz). For the CNN, HbO/HbR form 2 input channels over 420 time steps for each of the 36 channels |
| Label | 0 = LOW (0-back); 1 = HIGH (2-/3-back) |
| Subject ID | VP001–VP011, VP014–VP026 (24 subjects), used to build LOSO folds |
| Dataset size | 648 epochs = 24 subjects $\times$ 27 blocks (216 LOW, 432 HIGH) |
| Per-fold sets | Train: 22 subjects (594 epochs); validation: 1 subject (27); test: 1 subject (27) |
| Normalisation store | Channel-wise $\mu$, $\sigma$ per fold, from training subjects only |
| Outputs | Per-fold checkpoint, metrics table, confusion matrix, pooled predictions for all 648 test trials |

# Chapter 5 Methodology

## 5.1 Tools and Techniques

- **Paradigm:** N-back (0-, 2-, 3-back) as recorded in the TU Berlin dataset [16]; PsychoPy [10] planned for future data collection.
- **Signal processing:** band-pass filters, mean removal, common average referencing and baseline correction using MNE-Python [7] and SciPy; ICA [8] and the Modified Beer–Lambert Law [18] for raw-data pipelines.
- **Deep learning (PyTorch):** 1D CNN, bidirectional LSTM [19], temporal attention pooling [21], temporal CNN for fNIRS, fusion MLP.
- **Training:** class-weighted cross-entropy, Adam, per-fold dropout selection, ReduceLROnPlateau, early stopping.
- **Evaluation:** LOSO cross-validation; Macro-F1 (primary), Accuracy, Macro-Precision, Macro-Recall, ROC-AUC, confusion matrices (scikit-learn).

## 5.2 Data Set with Explanation

The TU Berlin simultaneous EEG–fNIRS dataset [16] is used. Twenty-four subjects (VP001–VP011, VP014–VP026) each perform 27 n-back blocks (9 of 0-back, 18 of 2-/3-back), giving 648 paired 42-second epochs: 216 LOW and 432 HIGH. Because HIGH is twice as frequent as LOW, Macro-F1 is the primary metric.

| Item | Value |
|---|---|
| Subjects | 24 (VP001–VP011, VP014–VP026) |
| Blocks per subject | 27 (9 $\times$ 0-back, 18 $\times$ 2/3-back) |
| Epochs | 648 paired 42-s epochs (216 LOW, 432 HIGH) |
| EEG | 28 channels, 200 Hz, 8,400 samples per epoch |
| fNIRS | 36 channels, 10 Hz, 420 samples, HbO and HbR |
| Labels | 0-back $\rightarrow$ LOW (0); 2- and 3-back $\rightarrow$ HIGH (1) |

Table 5.1: Dataset summary

## 5.3 Flowchart of Methodology from Start to End

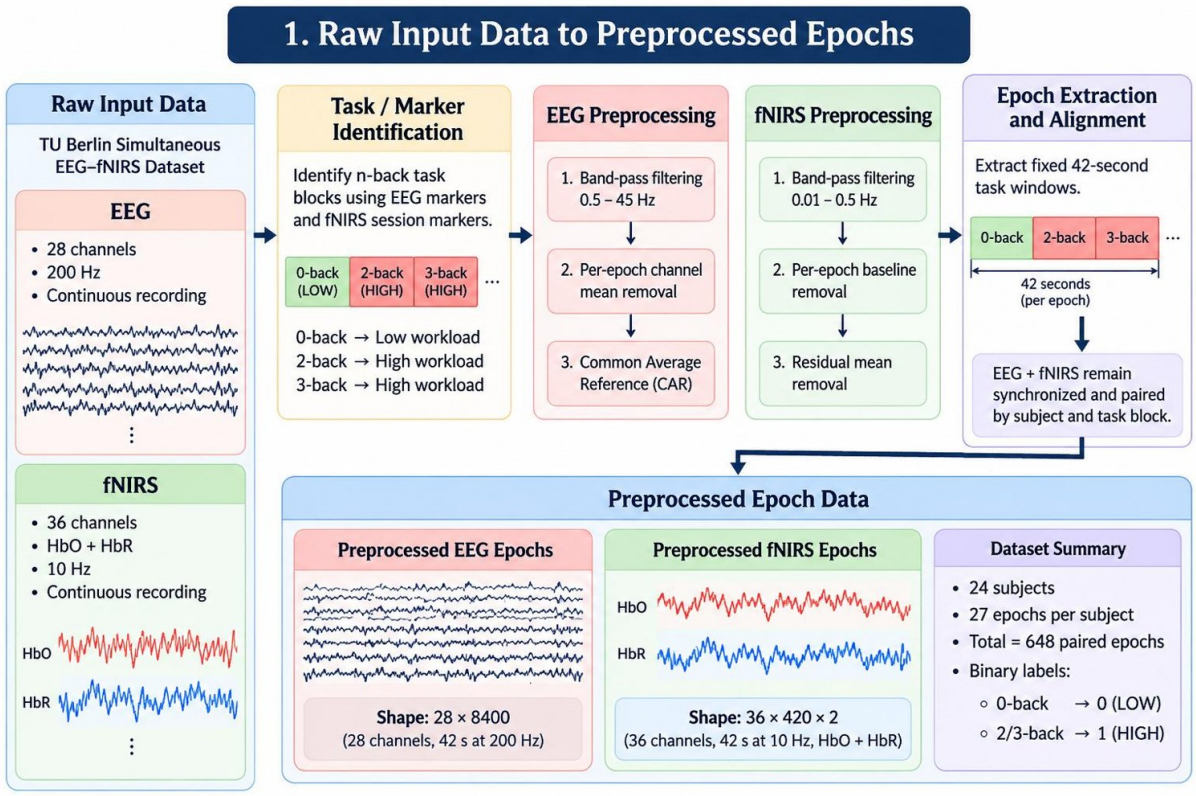



Figure 5.1: Raw input data to preprocessed epochs

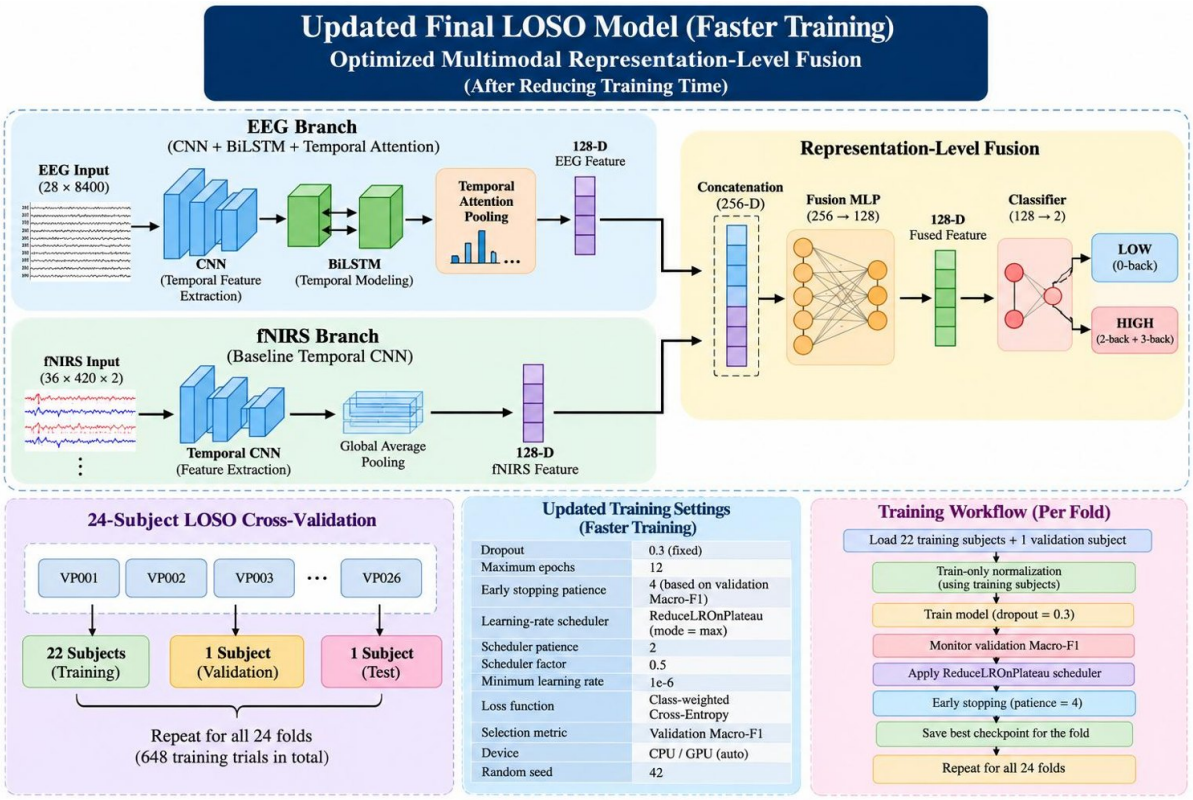



Figure 5.2: Final LOSO model: representation-level EEG–fNIRS fusion

The complete procedure, which can be followed step by step to reproduce the work, is:

1. **Obtain data.** Download the TU Berlin EEG–NIRS dataset [16]; keep subjects VP001–VP011 and VP014–VP026.

2. **Identify blocks.** Using the task markers, locate each n-back block (27 per subject) and its condition (0-, 2- or 3-back).

3. **Label.** 0-back $\rightarrow$ 0 (LOW); 2-/3-back $\rightarrow$ 1 (HIGH).

4. **Preprocess EEG.** Band-pass 0.5–45 Hz; subtract each channel's mean per epoch; apply common average reference. Result: $28 \times 8400$.

5. **Preprocess fNIRS.** Band-pass 0.01–0.5 Hz; remove baseline; remove residual mean for both HbO and HbR. Result: $36 \times 420 \times 2$.

6. **Epoch and pair.** Cut each block into a fixed 42-second window for both modalities so that EEG and fNIRS epochs of the same block stay paired. Total 648 pairs. No aggressive artefact rejection is applied.

7. **Create LOSO folds.** For fold $f = 1..24$: test subject = subject $f$; validation subject = another fixed subject (rotation rule from the code **[state the rule used]**); training = remaining 22.

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8. **Normalise.** Compute channel-wise $\mu, \sigma$ on training subjects only; apply to validation and test.
9. **Select dropout.** For each $p \in \{0.2, 0.3, 0.5\}$ train briefly and pick $p$ with best validation Macro-F1.
10. **Train.** Adam, class-weighted cross-entropy, ReduceLROnPlateau (factor 0.5, patience 2, min LR $10^{-6}$), at most 12 epochs, early stopping with patience 4 on validation Macro-F1; keep the best checkpoint.
11. **Test.** Load the best checkpoint and predict the held-out subject; store labels, probabilities and metrics.
12. **Repeat and aggregate.** Repeat for all 24 folds; report mean $\pm$ SD and pooled results over 648 test trials.
13. **Fusion variants.** Repeat with unimodal models, decision-level fusion (average of probabilities, thresholded, learned) and logistic fusion for comparison.

## 5.4 Explanation of Methodology in Detail with Equations and Formulae

### 5.4.1 Preprocessing

**Common average reference.** For EEG sample $x_c(t)$ at channel $c$ among $C = 28$ channels,

$$x_c^{\mathrm{CAR}}(t) = x_c(t) - \frac{1}{C}\sum_{j=1}^{C} x_j(t). \tag{5.1}$$

**Channel mean removal.** $x_c(t) \leftarrow x_c(t) - \frac{1}{T}\sum_t x_c(t)$, with $T = 8400$ (EEG).
**Band-pass.** EEG 0.5–45 Hz removes slow drift and line/muscle noise; fNIRS 0.01–0.5 Hz removes drift and cardiac/respiratory components while keeping the haemodynamic response.
**Baseline and residual-mean removal (fNIRS).** The baseline level of each channel and chromophore is subtracted so that only the task-related change remains.

### 5.4.2 Evaluation protocol and normalisation

In every fold, 22 subjects train, 1 validates and 1 tests; each subject is the test subject once. Normalisation, dropout selection, scheduling and checkpoint choice use only training and validation subjects. For channel $c$,

$$\hat{x}_c(t) = \frac{x_c(t) - \mu_c^{\mathrm{train}}}{\sigma_c^{\mathrm{train}}}. \tag{5.2}$$


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### 5.4.3 EEG branch: CNN–BiLSTM with temporal attention

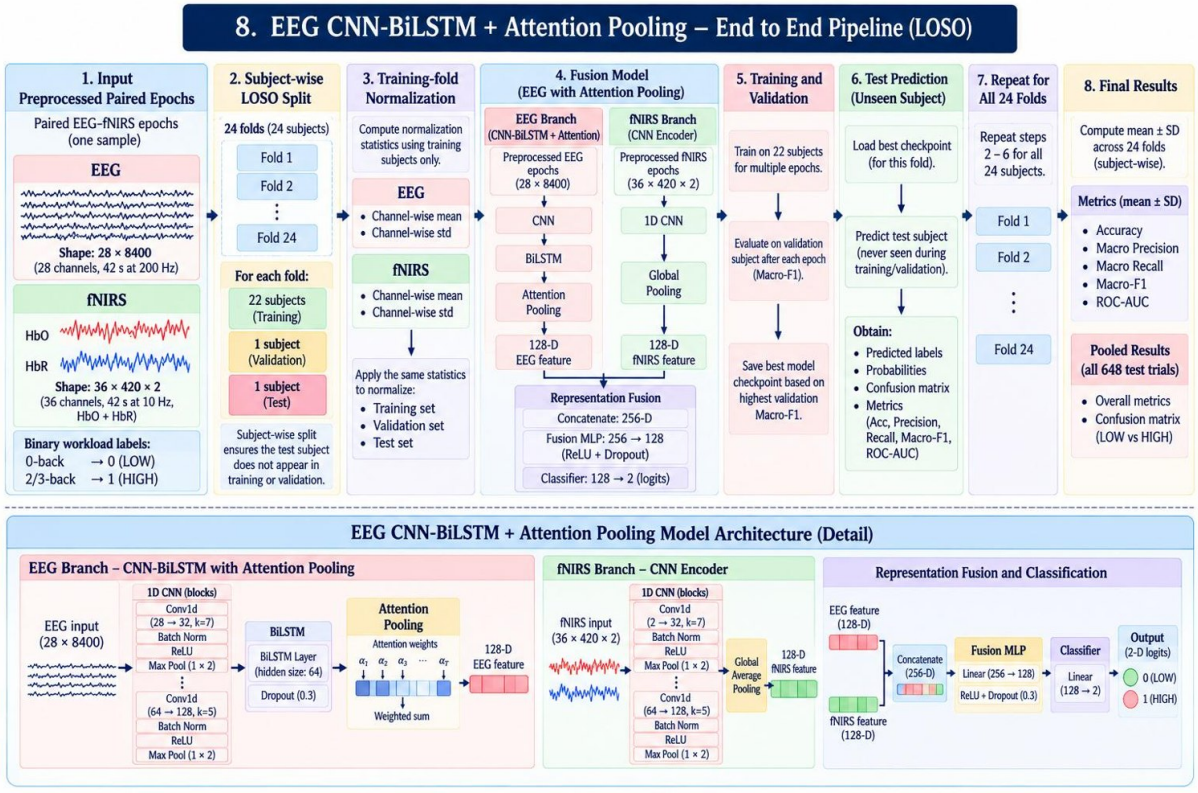



Figure 5.3: EEG branch: CNN-BiLSTM with temporal attention pooling

The $28 \times 8400$ epoch enters stacked Conv1D–BatchNorm–ReLU–MaxPool blocks (first block $28 \to 32$ channels, kernel 7; last block ends at 128 channels with kernel 5) **[verify intermediate layers against code]**. Pooling is kept mild so more of the 42-second window is retained. A 1D convolution computes

$$y_k(t) = \mathrm{ReLU}\Big( \sum_{c} \sum_{m=0}^{K-1} w_{k,c,m}\, x_c(t+m) + b_k \Big). \tag{5.3}$$

The CNN output sequence $\{u_t\}$ feeds a bidirectional LSTM (hidden size 64 per direction, so each $h_t \in \mathbb{R}^{128}$). Each LSTM direction uses

$$i_t = \sigma(W_i u_t + U_i h_{t-1} + b_i), \qquad f_t = \sigma(W_f u_t + U_f h_{t-1} + b_f), \tag{5.4}$$

$$o_t = \sigma(W_o u_t + U_o h_{t-1} + b_o), \qquad \tilde{c}_t = \tanh(W_c u_t + U_c h_{t-1} + b_c), \tag{5.5}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \qquad h_t = o_t \odot \tanh(c_t), \tag{5.6}$$

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and $h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}]$. **Temporal attention pooling** replaces mean/max pooling:

$$e_t = v^\top \tanh(W_a h_t + b_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_s \exp(e_s)}, \quad \mathbf{h}_{\text{EEG}} = \sum_t \alpha_t h_t, \tag{5.7}$$

giving the 128-D EEG representation. Attention lets the model weight the time positions most related to workload, since changes are not uniform across the epoch.

### 5.4.4 fNIRS branch: temporal CNN

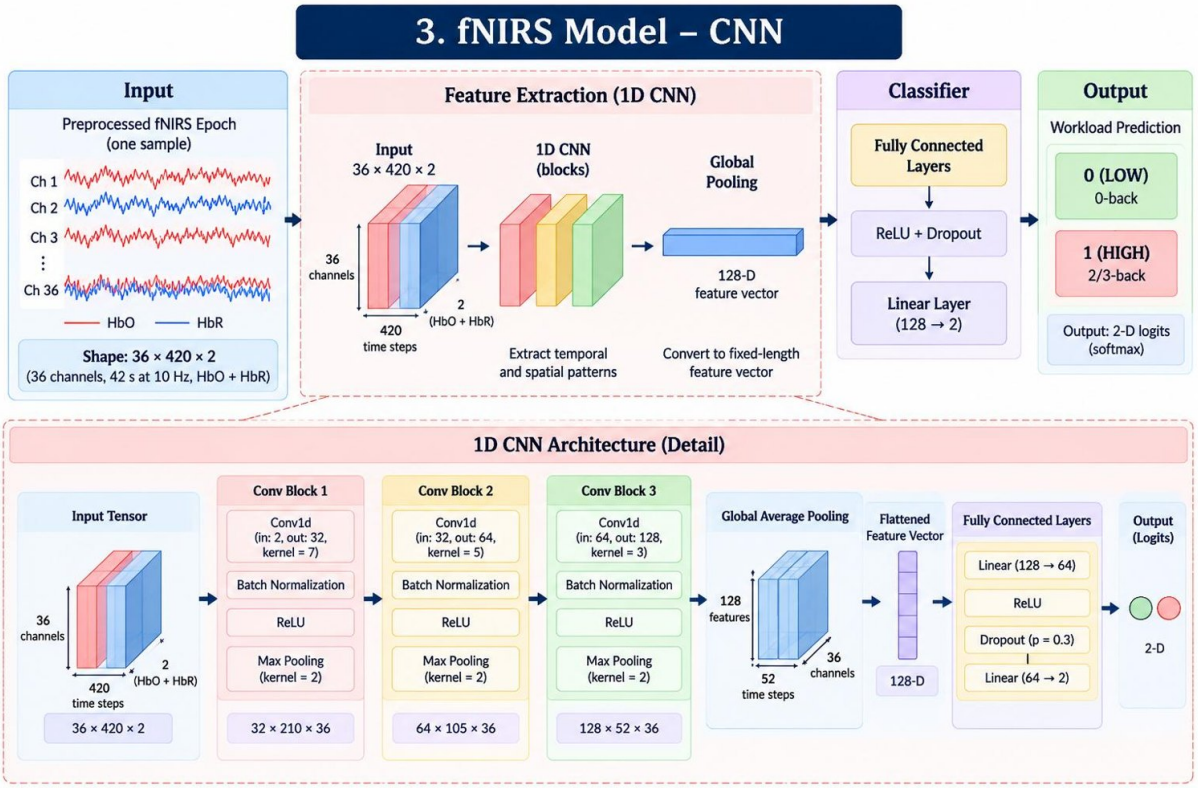



Figure 5.4: fNIRS branch: temporal CNN

HbO and HbR are two input channels. Three Conv1D blocks ($2 \to 32$, $k = 7$; $32 \to 64$, $k = 5$; $64 \to 128$, $k = 3$), each followed by BatchNorm, ReLU and MaxPool (kernel 2), extract slow haemodynamic patterns. Global average pooling over time gives $\mathbf{h}_{\text{fNIRS}} \in \mathbb{R}^{128}$. A deliberately simple CNN is used because fNIRS is slow and smooth and only 24 subjects are available, so larger models overfit.

### 5.4.5 Representation-level fusion and classifier

$$\mathbf{z} = [\mathbf{h}_{\text{EEG}}; \mathbf{h}_{\text{fNIRS}}] \in \mathbb{R}^{256}, \qquad \hat{\mathbf{y}} = \text{softmax}\big(W_2\,\text{Dropout}(\text{ReLU}(W_1\mathbf{z} + b_1)) + b_2\big), \tag{5.8}$$


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with $W_1 \in \mathbb{R}^{128\times256}$ and $W_2 \in \mathbb{R}^{2\times128}$. Fusing learned features before the decision lets the classifier use the fast electrical and slow haemodynamic information together.

### 5.4.6 Decision-level fusion (comparison)

The EEG and fNIRS models are trained separately and their class probabilities are averaged:

$$p(\text{HIGH}) = w\, p_{\text{EEG}} + (1-w)\, p_{\text{fNIRS}}, \qquad w = 0.5 \text{ (50:50 fusion).} \tag{5.9}$$

Variants tested: thresholded decision, learned dual-temporal-CNN fusion and logistic regression on frozen embeddings.

### 5.4.7 Loss and training

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i} \log \hat{y}_{i,y_i}, \tag{5.10}$$

where $w_k$ compensates the 1:2 LOW:HIGH ratio (inverse class frequency, e.g. $w_k = N/(2N_k)$ **[confirm exact weights]**). Settings are in Table 5.2.

| Setting | Value |
|---|---|
| Loss | Class-weighted cross-entropy |
| Optimiser | Adam (learning rate **[fill from code]**; batch size **[fill from code]**) |
| Dropout | Selected per fold from $\{0.2, 0.3, 0.5\}$ on validation Macro-F1 |
| LR scheduler | ReduceLROnPlateau on validation Macro-F1 (factor 0.5, patience 2, min LR $10^{-6}$) |
| Max epochs / early stopping | 12 / patience 4 on validation Macro-F1 |
| Checkpoint | Best validation Macro-F1 per fold |
| Normalisation | Channel-wise mean/std from training subjects only |
| Seed / device | 42 / CPU or GPU (automatic) |

Table 5.2: Final model training settings

### 5.4.8 Evaluation metrics

With $TP_k, FP_k, FN_k$ for class $k \in \{\text{LOW}, \text{HIGH}\}$:

$$\text{Accuracy} = \frac{\sum_k TP_k}{N}, \tag{5.11}$$

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \tag{5.12}$$

$$F1_k = \frac{2\,\text{Precision}_k \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}, \quad \text{Macro-F1} = \frac{1}{2}\sum_k F1_k. \tag{5.13}$$


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25 ROC-AUC is the area under the true-positive vs false-positive curve over all thresholds. A classifier that always predicts HIGH scores $432/648 = 66.7\%$ accuracy but Macro-F1 $= (0.80 + 0)/2 = 0.40$, which is why Macro-F1 is primary.

# Chapter 6 Results and Discussion

## 6.1 Presentation of Results

| Metric | Mean ± SD over 24 folds |
|---|---|
| Accuracy | 66.36% ± 15.48% |
| Macro-F1 (primary) | 57.70% ± 17.78% |
| ROC-AUC | 0.7842 ± 0.1502 |
| Macro-Precision (pooled) | 63.16% |
| Macro-Recall (pooled) | 62.15% |
| HIGH-class recall | 74.8% |
| LOW-class recall | 49.5% |

Table 6.1: 24-fold LOSO performance of the final model

The model reaches 66.36% accuracy and 0.7842 ROC-AUC on unseen subjects. Large standard deviations (15–18 points) reflect strong inter-subject variability typical of subject-independent EEG–fNIRS decoding. HIGH workload is recognised well (74.8% recall) but LOW less reliably (49.5%), because LOW is the minority class (216 of 648 epochs).

### 6.1.1 Comparison with reference models

| Model | Accuracy | Macro-F1 | ROC-AUC |
|---|---|---|---|
| EEG baseline (CNN–BiLSTM) | 60.80% | 52.16% | 0.7814 |
| fNIRS baseline (CNN) | 63.27% | 56.99% | 0.6407 |
| Baseline representation fusion | 64.51% | 59.12% | 0.7482 |
| Final model (EEG attention + fNIRS CNN) | 66.36% | 57.70% | 0.7842 |

Table 6.2: Final model versus unimodal and baseline fusion references

- Versus EEG alone, the final model gains 5.56 points of accuracy and 5.54 of Macro-F1; versus fNIRS alone, 3.09 and 0.71. Fusion therefore adds value over either modality.
- Versus baseline representation fusion, accuracy rises 1.85 points and ROC-AUC 0.036, but Macro-F1 is 1.42 points lower; given the 15–18 point SD this is within fold-to-fold variation.
- The temporal-attention EEG branch has a standalone ROC-AUC of 0.8138, above the plain CNN–BiLSTM (0.7814).

### 6.1.2 Best result among fusion variants

Simple 50:50 decision-level fusion obtained the best Macro-F1 among the fixed approaches (62.26%), above the optimised representation-level model (57.70%). Logistic fusion on frozen

| Configuration | Acc. | Macro-F1 | ROC-AUC |
|---|---|---|---|
| 50:50 decision-level fusion (best fixed) | 67.44% | 62.26% | 0.7521 |
| Thresholded decision-level (separate run) | 68.06% | 62.89% | n/a |
| Learned dual-temporal-CNN fusion | 66.82% | 61.18% | 0.7855 |
| Logistic embedding fusion | 67.59% | 61.16% | 0.7630 |
| Optimised representation fusion (final) | 66.36% | 57.70% | 0.7842 |

Table 6.3: Main fusion configurations (24-fold LOSO)

embeddings raised accuracy by 0.15 points and ROC-AUC by 0.0109 over 50:50 fusion but lowered Macro-F1 by 1.10 points, did not improve calibration and took about 68 minutes on a CPU, so it was kept as diagnostic evidence only. Thresholded fusion is a separate run, so its figures are shown as reported. The final model has the highest ROC-AUC except for the learned dual-temporal-CNN variant (0.7855), yet a lower Macro-F1, showing that better class ranking did not translate into better predictions at the default decision rule.

### 6.1.3 Comparison with existing literature

Published studies differ in dataset, task, classes and evaluation protocol, so accuracies cannot be set directly against ours. Table 2.2 and Table 6.4 summarise what differs. Our results agree with the main conclusion of [1, 2, 4, 23]: combining EEG and fNIRS improves on either modality alone.

| Aspect | Often seen in published studies | This work |
|---|---|---|
| Test subjects | Same subjects may be in training and test | Test subject never seen (LOSO) |
| Samples | Many short windows per subject | One 42-s epoch per block: 648 in total |
| Classes/metric | Balanced classes or task vs rest; accuracy | 1:2 imbalance; Macro-F1 primary; majority baseline 66.7% |
| Model selection | Sometimes on data related to test data | Separate validation subject |
| Artefacts | ICA and rejection common | Filtering and referencing only |

Table 6.4: Evaluation choices that affect comparability

### 6.1.4 Why our results differ from published results

- **Strict subject-independent evaluation.** A fixed 18/3/3 subject split gave 72.84% accuracy and 68.85% Macro-F1 for the same representation-fusion idea, against 64.51% and 59.12% in 24-fold LOSO (indicative only, three test subjects).

- **Inter-subject variability and small sample.** Per-fold SDs of roughly 12–22 points mean some subjects are near chance [15, 17]; 1–3 point differences lie within this variation.
- **Class imbalance and metric.** Always predicting HIGH gives 66.7% accuracy, so accuracy alone overstates performance.
- **Task definition.** 0-back vs 2-/3-back is a harder contrast than task vs rest [23, 24]; labels are conditions, not measured effort.
- **Epoch construction.** Whole 42-s blocks are kept together, avoiding the leakage of window-level splits.
- **Minimal artefact handling.** No ICA or motion correction [7, 8], consistent with the weak EEG-only Macro-F1 (52.16%).
- **Limited tuning on one validation subject.** Weight selection on one validation subject improved 9 folds and worsened 11 (mean Macro-F1 $-4.32$ points).
- **Modality characteristics.** EEG ranks classes better (AUC 0.78 vs 0.64), fNIRS is more stable in accuracy and Macro-F1; they are complementary [1, 2].

Our lower scores mainly reflect a stricter test, a small dataset, an imbalanced binary task and conservative preprocessing and tuning, not a failure of fusion. A like-for-like claim against a specific published method requires re-running it on this dataset under the same LOSO protocol (future work).

### 6.1.5 How the gap can be narrowed

Class-weighted unimodal encoders and balanced-accuracy model selection; validation-only threshold selection and calibration; hybrid band-power and HbO/HbR statistics; ICA and motion correction; transfer learning or short calibration blocks; multi-seed runs with paired tests across subjects.

## 6.2 Limitations of the Study

- High variance across subjects; differences of 1–2 points are within one standard deviation.
- Low LOW-class recall (49.5%) due to imbalance.
- Binary only; the three-state design is not implemented.
- Offline evaluation on a public dataset; real-time latency untested.
- Tuning on a single validation subject; training limited to 12 epochs.

## 6.3 Conclusion

Fusing EEG and fNIRS improved on either modality alone on unseen subjects, with the final model reaching 66.36% accuracy, 57.70% Macro-F1 and 0.7842 ROC-AUC, while simple 50:50 decision fusion gave the best Macro-F1 (62.26%). The absolute scores are modest because the evaluation is strict, and the main weakness is recall of the LOW class.

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# Chapter 7 Conclusion

## 7.1 Summary

This project built a multimodal EEG–fNIRS framework for cognitive workload assessment. The TU Berlin dataset was preprocessed into 648 paired 42-second epochs (EEG $28 \times 8400$, fNIRS $36 \times 420 \times 2$). The final model combines a CNN–BiLSTM encoder with temporal attention pooling for EEG and a temporal CNN for fNIRS, fused by an MLP, and was trained with class-weighted cross-entropy, per-fold dropout selection, ReduceLROnPlateau and early stopping under 24-fold LOSO evaluation.

## 7.2 Conclusion

- The final model achieved 66.36% accuracy, 57.70% Macro-F1 and 0.7842 ROC-AUC, improving on EEG alone or fNIRS alone and supporting multimodal fusion.
- Temporal attention in the EEG branch improved class ranking, and a simple CNN suited the slow fNIRS signal.
- HIGH workload is recognised well (recall 74.8%) while LOW workload is recognised less reliably (recall 49.5%).
- Subject variability is large (about $\pm$15–18 points), so 1–2 point improvements may not be significant.
- Simple equal-weight decision fusion gave the best Macro-F1 (62.26%).
- Scores are lower than some published results because of strict LOSO evaluation, a small dataset, class imbalance and conservative preprocessing and tuning; results cannot be compared directly with studies that use other tasks.
- The initial target of $F1 \geq 0.80$ was not achieved (about 0.58 under LOSO).

## 7.3 Future Work

- Improve LOW-class recall with class-weighted unimodal encoders and balanced-accuracy model selection.
- Tune the decision threshold and calibrate probabilities using validation data only.


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- Audit fNIRS channel quality, motion artefacts and epoch timing; add ICA and motion correction.
- Hybrid fusion of deep features with EEG band power and HbO/HbR statistics.
- Logistic meta-fusion using out-of-fold predictions to avoid leakage.
- Three workload states, real-time acquisition with PsychoPy once fNIRS hardware is available, plus dashboard and API integration.
- Re-run selected published methods on this dataset for a fair comparison, with multiple seeds and paired subject-level tests.

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# Plagiarism

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**[Attach the similarity report and the AI-generated content report here (Appendix F).]**

# Appendix A: Snapshots

**[Add screenshots of code, training logs and result plots. Design and architecture figures are in Chapter 4.]**

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# Appendix B: Details of List of Publications Related to this Project

Author names:

Paper title:

Conference/Journal:

Place of conference:

Date of conference:


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# Appendix C: Details of Patent

# Appendix D: Details of Funding

# Appendix E: POs and PSOs Mapped

Table E.1: Programme outcomes mapping

| PO | Level | Justification |
|---|---|---|
| PO1 | 3 | Applies engineering, signal-processing and deep-learning knowledge to EEG–fNIRS workload classification. |
| PO2 | 3 | Analyses limitations of unimodal sensing, artefacts, synchronisation and subject variability through a research-gap analysis. |
| PO3 | 3 | Designs a full multimodal pipeline: preprocessing, encoders, fusion, classification. |
| PO4 | 3 | Controlled 24-fold LOSO experiments with baselines and fusion variants, with analysis. |
| PO5 | 3 | Uses Python, PyTorch, MNE-Python, scikit-learn and PsychoPy. |
| PO6 | 2 | Supports workload and fatigue monitoring relevant to healthcare, aviation and education. |
| PO7 | 1 | Open datasets and software avoid extra hardware and cost. |
| PO8 | 2 | Leakage-free, transparent reporting and an openly licensed public dataset. |
| PO9 | 3 | Developed collaboratively by a four-member team. |
| PO10 | 2 | Clear documentation, figures and result tables. |
| PO11 | 2 | Completed at NIL cost using college and free cloud resources. |
| PO12 | 2 | Identifies future improvements supporting continued learning. |
| PSO1 | 1 | Minimal networking; API integration and real-time streaming are future work. |
| PSO2 | 3 | Applies software-engineering practices and AI to a real-world workload problem. |
| PSO3 | 3 | Integrates deep learning and information systems into a scalable, hardware-agnostic solution. |

# Appendix F: Similarity Report and AI-Generated Report

**[Attach reports here.]**